Operating Systems — Unit 1: Processes and CPU Scheduling

A process is a program in execution, with its own address space, program counter and registers, tracked by the kernel in a process control block (PCB). States: new, ready, running, waiting, terminated.

Scheduling algorithms

• FCFS: simple, but short jobs wait behind long ones (convoy effect).
• SJF / SRTF: optimal average waiting time, needs burst-time estimates.
• Round Robin: time quantum q; small q means more context switches.
• Priority scheduling: risk of starvation, fixed by ageing.